

CNG Cylinder Tracking — SAP Data Integration

How SAP should push machine & cylinder data to the JP Minda CNG platform

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1. Overview

SAP sends **one message per machine** — the machine's current reading plus the cylinder (Pipe ID) it is working on — to a single secure HTTPS endpoint as JSON (`POST`). We receive it, store it, and it appears live on the cylinder dashboard. You do not need to match a rigid schema; you map your SAP fields to the simple shape below.

2. Endpoint & authentication

Item	Value
URL	<code>POST https://digitisationapi.jpmgroup.co.in/api/v1/sap/ingest</code>
Method	<code>POST</code>
Auth header	<code>x-api-key: <the key Deepnap issues you></code>
Content-Type	<code>application/json</code>

3. Request body — one machine per message

```
{
  "machineId": "HARDNESS-01",
  "machineName": "Hardness Tester",
  "stageNo": 7,
  "pipeId": "PIPE-60-V48002",
  "timestamp": "2026-07-01T10:15:00Z",
  "data": {
    "hardness": 92,
    "result": "PASS",
    "status": "running"
  }
}
```

4. Field guide

Field	Required	Meaning
<code>machineId</code>	required	Unique code for the machine (any naming, kept consistent).
<code>pipeId</code>	required	The cylinder / Pipe ID this machine is working on right now . This is what links every reading to the correct cylinder.
<code>timestamp</code>	required	When the reading was taken — ISO-8601 (e.g. <code>2026-07-01T10:15:00Z</code>).
<code>stageNo</code>	recommended	Which of the 21 stages this machine feeds (1–21). If omitted, we map it from the machine.

machineName	optional	Friendly display name.
data	required	The actual readings — you choose the field names . Put pass/fail as <code>result: "PASS"/"FAIL"</code> and running state as <code>status: "running"/"idle"/"stopped"/"fault"</code> .

You decide the names inside data . Common ones (pressure, temperature, hardness, depth, flow, thickness...) are shown as gauges automatically; any other field is still stored and displayed. Send what your SAP has.

5. Rules that keep it clean

- **Always include pipeId** — it links the reading to the cylinder (no barcode scanning needed on our side).
- **One machine per message**. Send a separate POST for each machine's reading.
- Send whenever a value changes, or on a short interval (e.g. every 5–15 seconds per machine).
- Send over **HTTPS to the domain above** (not an IP address).

6. Responses

Code	Meaning
202 Accepted	Received and stored. This is success.
401 Unauthorized	Missing or wrong <code>x-api-key</code> .
400 Bad Request	Body was not valid JSON.

Any valid JSON with a `machineId` is stored — even unexpected fields. The line never stalls.

7. Quick test (one line)

```
curl -X POST https://digitisationapi.jpmgroupp.co.in/api/v1/sap/ingest \
  -H "x-api-key: YOUR_KEY" -H "Content-Type: application/json" \
  -d '{"machineId":"TEST-01","pipeId":"PIPE-TEST-1","timestamp":"2026-07-01T10:15:00Z","data":
{"pressure":301,"result":"PASS"}}'
```

A `202` with `{"ok":true}` confirms the integration works end-to-end.

8. Go-live checklist

- ☐ Deepnap issues you the `x-api-key` .
- ☐ You map your SAP fields to the shape in §3 and send a few test messages.
- ☐ We confirm they arrive (raw log + live on the dashboard).
- ☐ Share your server's outbound source IP so we can whitelist it (defence in depth).
- ☐ Switch all machines to live — done.